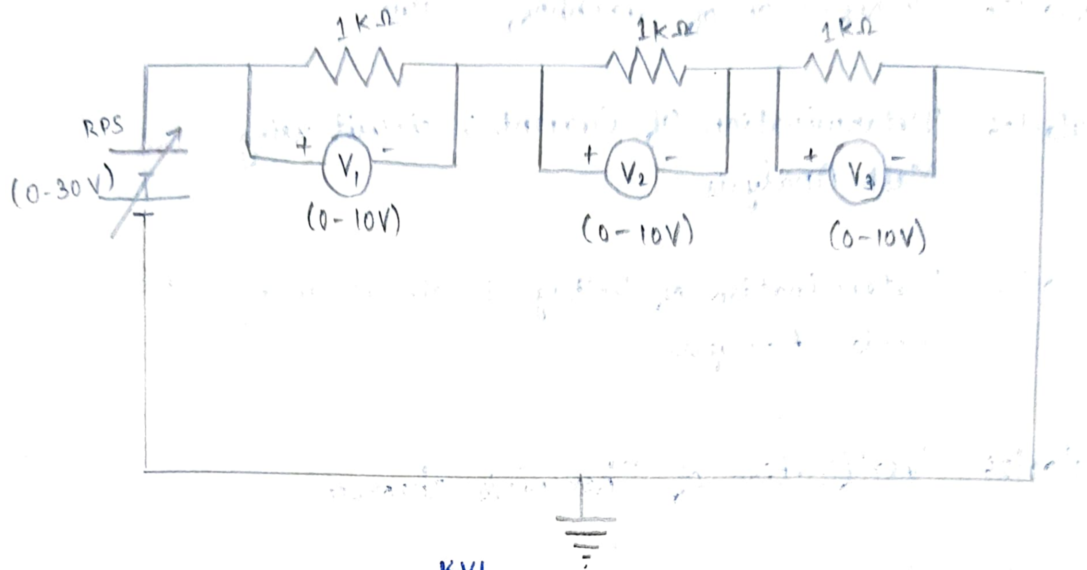
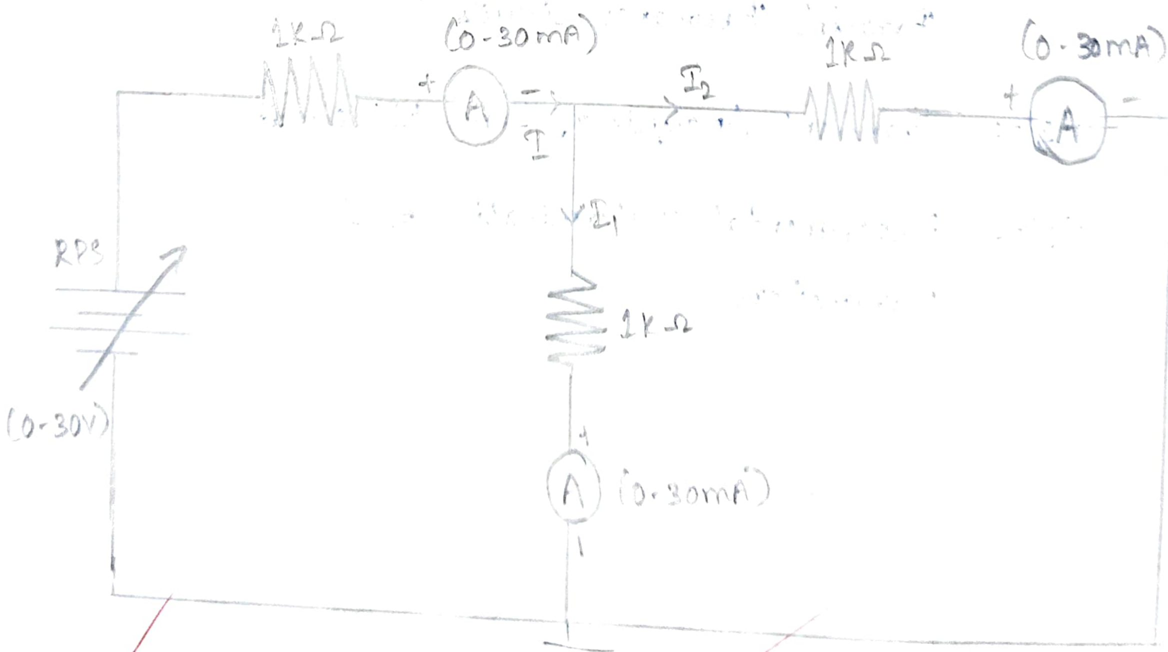
CIRCUIT DIAGRAM:



KVL



KCL

EXP NO. 1

3/2/26

VERIFICATION OF KIRCHHOFF'S LAWS

AIM:

To verify Kirchhoff's voltage law and Kirchhoff's current law both theoretically and practically for a given DC circuit.

APPARATUS REQUIRED:

SL.NO	APPARATUS	SPECIFICATION	QUANTITY
1.	Regulated Power Supply (RPS)	(0-30)V	1
2.	Voltmeter	(0-30V) MC	3
3.	Ammeter	(0-10mA) MC	3
4.	Resistors	1KΩ	3
5.	Bread board	-	1

PROCEDURE:

1. Give connections as per the circuit diagram.
2. Switch ON on the supply, vary the RPS (Regulated Power Supply) and set a particular input voltage.
3. Note down the readings of ammeters and voltmeters and tabulate them.
4. Vary the RPS for different input voltages and note down the readings of all the meters.
5. Reduce the RPS to its minimum value and switch OFF the supply.
6. Using the tabulated values, verify Kirchoff's laws practically, and verify it theoretically.

TABULAR COLUMN:

V (volts)	V_1 (volts)	V_2 (volts)	V_3 (volts)	$V = V_1 + V_2 + V_3$ (volts)	I_1 (amps)	I_1 (amps)	I_2 (amps)	$I = I_3 + I_2$ (amps)
3	1	1.03	1.04	3.07	2.03	1.02	1.03	2.05
6	1.95	2.07	6.05	10.17	4.02	2.01	2.02	4.03
9	2.89	3.08	8.97	14.94	6.03	3.01	3.02	6.03

Verified

THEORETICAL CALCULATION:

For V = 3V

$R_{eq} = 1+1+1 = 3k\Omega$

By Ohm's Law, $V = IR$

$$I = \frac{V}{R_{eq}} = \frac{3}{3k\Omega} = 1mA$$

$V_1 = R_1 I$

$V_1 = 1\times10^3 \times 1\times10^{-3} = 1V$

$V_2 = R_2 I \Rightarrow 1\times10^3 \times 1\times10^{-3} = 1V$

$V_3 = R_3 I \Rightarrow 1\times10^3 \times 1\times10^{-3} = 1V$

$V = V_1 + V_2 + V_3 = 1V + 1V + 1V = 3V$

For 6V,

$R_{eq} = 3k\Omega$

$I = \frac{V}{R} = \frac{6}{3} = 2mA$

$V_1 = R_1 I = 2V$

$V_2 = R_2 I = 2V$

$V_3 = R_3 I = 2V$

For V = 9V

$R_{eq} = 3k\Omega$

$I = \frac{V}{R} = \frac{9V}{3k\Omega} = 3mA$

$V_1 = R_1 I = 3V$

$V_2 = R_2 I = 3V$

$V_3 = R_3 I = 3V$

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For 3V,

$I = \frac{V}{R_1} = 2mA$, $R_{eq} = \frac{1\times1}{1+1} = 0.5k\Omega = 1.5k\Omega$

$I_1 = I \times \frac{1k\Omega}{1+1k} = \frac{1\times10^3}{2\times10^3} \times 2\times10^{-3} = 1.0mA$

$I_2 = I \times \frac{1k\Omega}{1+1k} = \frac{1\times10^3}{2\times10^3} \times 2\times10^{-3} = 1.0mA$

For 6V,

$I = \frac{V}{R} = 4mA$, $R_{eq} = \frac{1\times1}{1+1} = \frac{1}{2} = 2\times10^{-3} \times \frac{1\times10^3}{2\times10^3} = 1mA$

$I_1 = I \times \frac{1k\Omega}{1+1k} = \frac{1\times10^3}{2\times10^3} \times 4\times10^{-3} = 2mA$

$I_2 = I \times \frac{1k\Omega}{1+1k} = \frac{1\times10^3}{2\times10^3} \times 4\times10^{-3} = 2mA$

For 9V,

$I = \frac{V}{R} = 6mA$

$I_1 = I \times \frac{1k\Omega}{1+1k} = \frac{1\times10^3}{2\times10^3} \times 6\times10^{-3} = 3mA$

$I_2 = I \times \frac{1k\Omega}{1+1k} = \frac{1\times10^3}{2\times10^3} \times 6\times10^{-3} = 3mA$

$R_{eq} = \frac{1\times1}{1+1} = 0.5k\Omega = 0.5+1 = 1.5k\Omega$

S. NO	DESCRIPTION	MARKS ALLOTTED	MARKS OBTAINED
1.	Experimental setup/connections	20	20
2.	Observation / Readings	20	20
3.	Calculations / Results	20	20
4.	Viva	20	20
5.	Record	20	20
		Total :	100

100

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RESULT:

Thus, Kirchoff's current law and Kirchoff's Voltage law are verified practically and theoretically.